A Comprehensive Study of IoT-Based Assistive Stick for Elderly Safety and Mobility Support

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***Abstract*—The growing elderly population has led to an increase in mobility-related risks, particularly for individuals with visual or physical limitations. Many of these accidents become severe or even fatal because help does not reach the person in time. The delay in notifying family members, caregivers, or medical responders after a fall or emergency is often the main reason necessary assistance arrives too late.**

With the rise of intelligent systems built using the Internet of Things (IoT), Artificial Intelligence, and sensor-driven automation, a Smart Walking Stick can offer a dependable solution to this problem. The system designed in this work integrates Ultrasonic sensors, GPS, and GSM communication to continuously track the user's surroundings and movement behavior.

This paper presents an affordable, lightweight, and effective Smart Walking Stick model that uses IoT-based sensing and communication technologies to enable quick emergency alerts. By detecting obstacles, falls, and user location in real time, the proposed system enhances safety, supports independent mobility, and significantly reduces the chances of delayed emergency assistance.

***Index Terms*—Internet of Things (IoT), Sensor-driven automation, Smart Walking Stick, Ultrasonic sensors, independent mobility, Emergency Assistance.**

I. Introduction

As people start ageing, their bodily functions experience gradual changes in their physical strength, balance, power of sight and coordination that tend to change or reduce over time. These changes can make daily activities challenging to perform and can cause an increase in the chances of unfortunate incidents. Global population studies call attention to a substantial growth in the number of older adults, as the World Health Organisation reported in 2023 that by 2050, people aged 60 and above will constitute nearly one-sixth of the world's population. With this steady increase in the senior citizens, there is a burgeoning demand for assistive technologies that help in enhancing mobility, offer immediate support during emergencies, and maintain continuous safeguarding.

In developing countries and various parts of the world like India, Golden ager often manage their daily life and activities with partial guidance. Due to various reasons like urbanisation, the shift from joint families to nuclear families, and hectic work practices has reduced the amount of usual care taken towards seniors and older parents. These are some of the reasons that make them more vulnerable

to circumstances such as falls, disorientation, delays in immediate medical support, and several mobility challenges. The consequences of delayed or a lack of assistance can be serious. According to the World Health Organisation's 2024 report, falls rank as the second most common cause of unintentional injury-related deaths worldwide, highlighting the need for prompt detection and timely intervention.

Walking sticks have been traditionally used for ages and served as a simple walking cane, which was basically used by elderly people for basic support, balance, stability and mobility. However, these standard traditional devices are limited to just physical assistance and cannot sense dangerous situations, ask for help or communicate during critical times. Recent advancements in embedded systems, Internet of Things (IoT) areas, and low-power wireless communication technologies enable and open the door to opportunities in the development of intelligent mobility aids that exceed the expectations and capabilities of a standard walking cane. By integrating sensors and connectivity modules, a smart walking stick can continuously track user movement patterns, detect abnormalities or falls, trigger automated SOS alerts, help in medical support and send real-time location to caregivers or emergency first responders.

The presented IoT-enabled Assistive Stick for Elderly Safety and Mobility is designed in such a way that it helps in addressing the increasing need for intelligent support systems for senior citizens. The device we proposed incorporates a set of embedded hardware modules, in addition to an Arduino UNO controller, a GSM communication module, a GPS receiver, an emergency (SOS) trigger or alert, a buzzer, and a visual indicator such as LEDs. By combining and integrating these components, the system is capable of detecting emergency conditions, transmitting automated quick SMS notifications and alerts, providing live location data, and overall helping in improving the mobility and safety of older people.

II. Purpose of the Work

The main objective of this work is to design and analyse an IoT authorized aiding walking cane that helps in improving safety, mobility, and emergency response for elderly or senior users. Since falls are one of the most frequent and critical threats encountered by older adults, there is a clear necessity for a system that is capable of constantly keeping a check on user movements, identifying irregular or risky movement sequences, and immediately alerting caregivers and surroundings when an emergency occurs. Such a device can significantly aid independent living and help minimise delays in receiving timely assistance and help.

The study originally concentrates on developing a genuine method to analyse and interpret fall-related issues using a consolidation of sensors integrated into the walking cane. These sensors include accelerometers, pressure and barometric modules, and radar or ambient sensing units. Prior work, such as radar-assisted fall recognition [13], Doppler-based motion analysis [22], and multimodal monitoring frameworks for assisted living [15]demonstrates that precise signal interpretation is essential for identifying sudden events. Building on these findings, the presented system is designed to achieve stable, real-time monitoring regardless of whether the user is outdoors, indoors or moving over diversified surface conditions.

Another aim of this work is to estimate how well a portable, user-carried assistive tool can detect falls in comparison to fixed installations or fully wearable monitoring and assistive systems. In contrast to floor-mounted sensing frameworks or multi-node wearable setups [1], [2], [4], the walking stick integrates motion sensing, pressure variation analysis, and characteristic movement patterns with built-in emergency alert capabilities. The research further inspects the benefits of employing a lightweight, energy-aware sensing approach, particularly for older people who rely on the tool or mechanism for extended daily use. Power restraints remain a frequent issue in mobile assistive technologies, and prior work on enhanced sensing strategies and GPS efficiency [15] emphasises the importance and need for potential power management. In response to these challenges, the proposed assistive stick incorporates low-power hardware and controlled sensing intervals, ensuring that the mechanism remains practical and genuine throughout sustained operation.

Another important goal is to assess the trustworthiness of the system's emergency communication procedure, which is an important feature. When an unusual incident or potential fall is found, the stick instantly sends alerts that include the user's event time, location, and a basic categorisation of the incident. Earlier studies on smartphone-based fall detection [22] and IoT-operated alert mechanisms [21] illustrate the importance of such communication methods. This examination looks into whether a dedicated walking stick can provide faster and more consistent alerts, especially for older people who may not regularly, usually, carry or interact with a smartphone.

III. Motivation

Older people usually face challenges or issues such as reduced balance, stability, delayed response time, and various health-related challenges, all of which increase their vulnerability to falls and sudden medical issues. Conventional walking sticks offer only mechanical support and are unable to notify caretakers when an emergency takes place. According to WHO reports, falls continue to be a major factor that contributes most to injury-related fatalities among the elderly community [4]. This disadvantage in conventional mobility aids calls attention to the need for smarter, sensor-integrated answers capable of providing immediate and quick response, which helps in improving both safety and independence for older users.

IV. Contribution to the Work

This survey reviews existing work on IoT-based assistive sticks and groups past studies according to their sensing methods, alert mechanisms, and mobility-support features that are important for daily use [4]. It compares how different designs use components such as accelerometers, ultrasonic sensors, GPS/GSM modules, and microcontrollers, allowing a clear look at how well each system handles fall detection, obstacle sensing, and energy use in different settings.

It also examines how these devices perform in real-life situations—like changes in walking speed, uneven surfaces, different ways users hold the stick, and low-light conditions—to see how consistently they send alerts and keep communication links active under practical challenges [7]. The survey points out common issues, including limited fall-detection accuracy, slow alert responses, and a lack of strong multi-sensor integration. It also notes recent improvements that aim to make these systems more accurate and dependable while remaining low-cost and easy to carry.

In addition, the review gathers useful benchmarks such as detection accuracy, alert timing, sensor response, and user feedback. These insights can help developers, caregivers, and healthcare organisations choose reliable smart walking-stick systems and design future versions that are safer, more affordable, and better suited for real-world use.

V. Survey

1. Fall Detection Using Smartphone Audio Features

This paper presents a fall-detection method that uses audio picked up by a nearby smartphone. Using features such as spectrograms and MFCCs, an ANN provided the best performance, achieving over 98% accuracy, sensitivity, and specificity. [30]

2. Body Sensor Networks: In the Era of Big Data and Beyond

This paper discusses how sensors worn on the body can gather health data in real time and support medical monitoring. It also points out challenges, such as creating sensors that function well on or inside the body and managing the large amounts of data these systems generate. [29]

3. Developing Walking Assistants for Visually Impaired People: A Review

This survey examines modern technologies that have been created to help visually impaired individuals navigate safely. It evaluates different sensor-based, computer-vision, and smartphone-assisted systems, noting their advantages and disadvantages. This review also identifies major challenges and suggests future improvements for developing more reliable mobility assistance tools. [28]

4. Modelling and Improving the Energy Performance of GPS Receivers for Location Services

This paper analysed how GPS energy use scales with the number of tracked satellites and signal length. The authors proposed a selective-tracking algorithm that reduces power consumption while maintaining good positioning accuracy. [27]

5. A patient-specific single sensor IoT-based wearable fall prediction and detection system

This paper gives a patient-specific fall prediction and detection prototype system. It has a single tri-axial accelerometer attached to the patient's thigh to distinguish between activities of daily living and fall events. It consists of fast mode and slow mode for fall prediction. The nonlinear support vector machine classifier-based FMFP algorithm is used to extract seven discriminating features for the pre-fall case, to identify a fall risk event and alert the patient. The SMFD algorithm is used to identify a single and optimal threshold for each patient. FMFP achieves sensitivity and specificity of 97.8% and 99.1%, respectively. [26]

6. Design and Implementation of an Intelligent Assistive System for Visually Impaired People for Aerial Obstacle Avoidance and Fall Detection

This work presents a setup using smart glasses and a smart walking stick to help visually impaired users detect obstacles and falls, with alerts sent to caregivers when needed. The system detects obstacles within 3 meters and reaches 98.3% fall-detection accuracy. [25]

7. A Wearable Device for Indoor Imminent Danger Detection and Avoidance With Region-Based Ground Segmentation

This paper estimates a wearable device to help visually impaired people with their daily lives. The system has a Red, Green, Blue, and Depth (RGB-D) camera and an Inertial Measurement Unit (IMU), which detects objects and the clash-free path in real-time. With the growth of wearable devices, Internet of Things (IoT), and artificial intelligence (AI), smart walking sticks have emerged as an acceptable solution to enhance self-reliance and safety for users in both inside and outside environments. [23]

8. Elderly Fall Detection with Vital Signs Monitoring Using CW Doppler Radar

This study introduces a low-cost fall-detection system built on a 24 GHz CW Doppler radar capable of tracking indoor human motion even through obstacles and under any lighting conditions. The radar data were processed using feature engineering and machine learning to extract unique, nontraditional motion features. Additionally, vital sign monitoring was integrated to reduce false alarms and enable timely alerts. The experimental results reported a 90% fall detection recall and 97.7% and 95.3% accuracy for respiration and heartbeat detection, respectively. [18]

9. Human Fall Detection in Surveillance Videos Using Fall Motion Vector Modelling

This paper introduces a method that captures key spatio-temporal motion characteristics of the human body to achieve accurate fall detection. The approach constructs a Fall Motion Mixture Model (FMMM) using optical-flow and motion-boundary features, followed by factor analysis to generate a compact fall motion vector. Experiments on the Le2i, URFall, and Montreal datasets show that this method surpasses the performance of several existing fall-detection techniques. [22]

10. Development of Converged Device-Based Exercise Program for Preventing Fall and Increasing Physical Activity for the Elderly

This study introduces an easy-to-follow, device-supported home exercise program featuring eight movements tailored to reduce fall risk in frail older adults. The setup includes interaction between the user and a trainer and helps seniors become more familiar with digital devices. [21]

11. Internet of Things and Deep Learning Enabled Elderly Fall Detection Model for Smart Homecare

This study presents an IoT-powered system that uses an enhanced deep learning model to spot falls from indoor video streams. Once the video is processed, the system detects falls and immediately notifies caregivers. Tests on two datasets showed excellent performance, with accuracy reaching nearly 99.8%. [20]

12. Smart Wristband–Based Stress Detection Framework for Older Adults Using Cortisol as a Biomarker

This work presents a wristband system that uses multiple physiological signals to distinguish stress in older adults, validated against cortisol levels. In tests with 40 participants, the model performed well, reaching 94% accuracy. [19]

13. IoT-Enabled Smart Healthcare Monitoring System for Elderly People

This paper presents a system that uses IoT sensors to monitor an elderly person's vital signs and movements. The healthcare monitoring tackecare of the elderly one and ensures safety and security. The information is stored in the cloud so caregivers can check it from anywhere. It also gives alerts when something unusual is detected to help ensure safety at home. [3]

14. Detecting Elderly Behaviours Based on Deep Learning for Healthcare: Recent Advances and Challenges

This paper reviews how deep learning is being used in smart devices to monitor elderly behaviour, with CNNs being the most common approach. It studies recent advancements and challenges in the healthcare sector. It also highlights current challenges and suggests areas where future research is needed. [17]

15. AIoT-Based Smart Stick for Visually Impaired Persons

In this paper, they used AIoT to design Smart devices for the Visually Impaired Persons. The Developer proposed a smart stick that reduces most of the problems in daily life and gives mobility support for the elderly. It includes ultrasonic sensors, GPS, haptic feedback, and AI-driven obstacle detection for better navigation and safety. The smart stick is embedded with a Raspberry Pi and a global positioning system, which helps in sensing emergency alerts and coordinates. This came up with a good outcome. The Overall obstacle detection accuracy of the system is 91.7%. [16]

16. A Smart Walking Stick for Gait Analysis of the Elderly and People With Disabilities

In this paper, they developed a smart walking stick fitted with Accelerometers and pressure sensors to monitor gait parameters such as body movements, step length, and pressure distribution. The Gait Analysis is also a technique to track the person, because it will track the body movements, step length and all. The outcomes reveal that the accuracy of the number of steps can reach 100%. The average error of stride duration-0.01±0.07s, step length-0.72±5.22cm and step speed-0.75±3.74cm/s with an average accuracy of 96.43%, 95.20% and 94.45%, respectively. [15]

17. Smart Assistive System for Visually Impaired People Obstruction Avoidance Through Object Detection and Classification

This paper introduced a Smart system through object detection and classification methods. The importance of smart walking sticks resides in their capacity to hold modern technologies, like ultrasonic sensors, computer vision, artificial intelligence (AI), and haptic feedback to provide real-time obstacle detection and aid in navigation. It utilises Raspberry Pi 4B, Camera, Ultrasonic Sensor and Arduino. They also used Viola-Jones and the TensorFlow Object Detection algorithm. The system plays a vital role in elderly people. The device comprises features like GPS-based navigation, voice assistance, and emergency alerts, making it a powerful tool for increasing the quality of life for visually impaired individuals. [14]

18. A novel feature extraction method for pre-impact fall detection system using deep learning and wearable sensors

In this work, they presented a deep-learning-based pre-pressure fall detection system that detects a fall within 0.5 s of the fall initiation phase, thus providing enough lead time of 0.5 s. They used an automatic feature extraction methodology to extract temporal features from wearable sensors. It includes a deep neural classifier based on the collection of convolutional neural networks (CNN) and long short-term memory (LSTM) networks. The classifier has performed incredibly well in detecting the Fall with a sensitivity of 99.24% and an F1-score of 98.79%. [13]

19. Design and Implementation of an Intelligent Assistive System for Visually Impaired People for Aerial Obstacle Avoidance and Fall Detection

This paper introduces a smart assistive system designed for visually impaired individuals, by wearable smart glasses and an intelligent walking stick. The system includes the smart glasses, the walking stick, a mobile app, and a cloud-based platform for managing information. The tests show that the system can detect obstacles within 3 meters and is pretty accurate in detecting falls, like 98.3% accurate. The significance of this paper is that it could help make things more accessible and reduce the need for human help by using technology to create a supportive and respectful environment where every individual feels valued. [24]

20. Smart Home Devices for Supporting Older Adults: A Systematic Review

The review shows that only a small number of studies have tested smart home devices with older adults, and evidence of their real-life benefits remains limited. Most existing research centers on system performance, highlighting the need for more thorough, long-term studies. [12]

21. Ethical Issues in Near-Future Socially Supportive Smart Assistants for Older Adults

This study looks at the ethical challenges tied to smart assistants designed to help older adults with everyday tasks and social interactions. While these systems can lighten cognitive demands and support independence, they may misinterpret social situations and create concerns about privacy, autonomy, and trust. The authors emphasise the need for careful monitoring to ensure these tools remain safe and beneficial. [11]

22. Remote Health Monitoring Systems for Elderly People: A Survey

This paper looks at various technologies used to monitor elderly people in their homes. It describes how sensors and wearable devices check vital signs and daily movements to spot health issues or emergencies. It also points out common concerns such as privacy, battery life, and the need for stable connections. [10]

23. VisiSafe: Real-Time Fall Detection for Visually Impaired People Using RF Sensing

The system uses a UWB XeThru radar to spot falls by detecting deviations from normal movements, making it effective even for visually impaired users. It achieved an accuracy of 98.8% within a 1.5-meter indoor range. [9]

24. Real-Time Human Tracking Using Multi-Features Visual with CNN-LSTM and Q-Learning

This study enhances human tracking by combining facial and body features, allowing the system to continue tracking even when the face is unclear or occluded. The CNN–LSTM model handles multi-feature recognition, whereas Q-learning improves decision-making in changing environments. Several features are involved, like CNN-LSTM and Q-Learning, which make the model more effective. The system demonstrated strong real-time performance, achieving an accuracy of 91.5%. [8]

25. Fall Detection Method Based on a Human Electrostatic Field and VMD-ECANet Architecture

This study uses electrostatic field signals combined with a VMD-ECANet model to detect falls in a fully non-contact manner, reaching 96.44% accuracy. It provides an affordable and privacy-friendly solution for monitoring seniors who live alone. [7]

26. Artificial Intelligence and IoT in Elderly Fall Prevention: A Review

This review studies various research, surveys and experiments on elderly fall prevention applying AI and other technologies, like the Internet of Things (IoT), sensors, RADAR, IR, and hardware technologies. These devices can observe gait patterns, detect imbalance, and even predict falls before they occur (Mohan et al., 2024). It has been explored that in real-time AI–IoT technology will be a better solution for fall prevention in the elderly. [6]

27. Real-Time Fall Detection Using Wideband Radar and a Lightweight Deep Learning Network

In this work, a radar-based fall detection scheme that estimates data streaming and lightweight networks is proposed to boost the accuracy of fall detection. A lightweight network is developed to detect falls in real time. Moreover, training samples with various radar heights and distances are expanded to establish a multi-indoor-site behaviour dataset. when the proposed network has a small size of 1.9178 MB, the System acquired an accuracy of 0.9963 in a new scene. [5]

28. A novel embedded deep learning wearable sensor for fall detection

This work establishes a novel deep-learning model specially enhanced for edge devices capable of detecting falls. The wearable sensor embodies a pressure sensor, a three-axis gyroscope, and a three-axis accelerometer. The developed system works in real time and has a dual objective. Mainly, 2 types of datasets are used: self-collected and publicly available datasets (SisFall). Results obtained an accuracy of 99.38% and a deduction time of 25 ms. [4]

29. Unobtrusive Sleep Health Assessment Using Impulse Radar: A Pilot Study in Older People

The paper gives sleep health data, and this study tested a UWB impulse radar to monitor sleep in older adults, including those with early Alzheimer's. The system tracked movement, breathing, and heart rate well and reached about 74% accuracy in identifying sleep stages. The findings suggest it could be a useful, nonintrusive option for home-based sleep monitoring. [1]

TABLE I: SUMMARY OF EXISTING WORKS IN ELDERLY ASSISTIVE & FALL DETECTION SYSTEMS

No.	Platform	Evaluation Metrics	Acc.	Future Enhancement
1	Audio (MFCC)	Fall via sound detection [30]	98%	Noise handling
2	Body Sensors	Medical monitoring & data [29]	N/R	Low-power design
3	Review	Assistive tech for visually impaired [28]	N/R	Better robustness
4	GPS Algo	Low-power GPS performance [27]	Good	Optimization
5	Accel & SVM	Fall prediction & detection [26]	97–99%	Personalization
6	Glasses & Stick	Obstacle & fall detect [25]	98.3%	Miniaturization
7	RGB-D & IMU	Wearable danger detection [23]	N/R	Faster processing
8	Radar,ML	Fall & vital signs [10]	90–97%	Real-world tests
9	Video (FMMM)	Fall detection in videos [11]	High	Robustness
10	Exercise System	Fall-risk reduction program [12]	N/R	Large trials
11	IoT & DL	Smart-home fall detection [25]	99.8%	Deployment
12	Wristband	Stress detection [13]	94%	Broader testing
13	IoT Sensors	Vital signs & movement [14]	N/R	Reduce latency
14	Review (DL)	Elderly behaviour monitoring [15]	N/R	Benchmark datasets
15	AIoT Stick	Obstacle detection [16]	91.7%	Weight reduction
16	Smart Stick	Gait analysis [17]	96–100%	Usability tests
17	Pi & Camera	Object detection stick [2]	N/R	Accuracy evaluation
18	Wearable & DL	Pre-impact fall detection [19]	99.2%	Edge deployment
19	Glasses & Stick	Obstacle & fall detect [20]	98.3%	Reduce latency
20	Review	Smart-home devices [21]	N/R	Long-term studies
21	Ethics Study	Socially supportive assistants [22]	N/R	Secure design
22	Review	Remote elderly monitoring [18]	N/R	Battery & privacy
23	UWB Radar	Fall detection [23]	98.8%	Improve range
24	CNN–LSTM &Q-Learning	Human tracking [25]	91.5%	Low-light accuracy
25	Galvanic & DL	Non-contact fall detection [26]	96.4%	Noise robustness
26	Review (AIoT)	Fall prevention summary [27]	N/R	Real-time edge
27	Radar & DL	Real-time fall detection [28]	99.6%	Diverse subjects
28	Wearable DL	Real-time fall detect (edge) [29]	99.38%	Low-power design
29	UWB Radar	Sleep monitoring [1]	74%	Accuracy improvement

VI. METHODOLOGY

The proposed Approach, the EldCare Assistive Stick, is designed to give independent life and mobility support to elderly people. The System will detect emergencies and send an alert to the caregiver's phone with a short message and GPS coordinates. It provides live or on-demand location tracking. This helps avoid obstacles by using an ultrasonic sensor and has audible/LED indicators. It is also a battery-powered, wearable/portable, safe and easy to use.

It is conceptualised to locate emergencies, such as falls in real time, object detection-early fall prevention, and immediately alert caregivers. The process is activated when the elderly users interact with the smart walking stick. The stick is equipped with various sensors to gather real-time movement and environmental data.

In the first stage, the ultrasonic sensor will detect obstacles, and if any are detected, then the GPS will track and record the location. After that, GSM will send the Alert Message to the Caregivers or the Registered Person's Mobile. Finally, coordinating on location and fall detection.

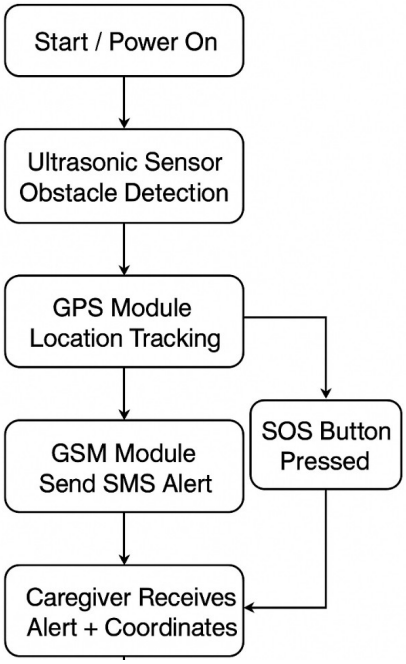

Fig. 1: Flow Diagram of EldCare Assistive Stick

- Normal mode: Stick powered on — GPS gets a fix; Ultrasonic sensor scans ahead for obstacles (vibrate/buzzer if object ¡ threshold); Periodic low-power GPS updates.
- SOS button pressed: Immediate SMS sent to predefined caregiver number with text and GPS link (Google Maps).

- Automatic fall detection: MPU6050 continuously monitors acceleration. If a sudden high-magnitude acceleration followed by near-zero motion is detected (threshold), the Arduino triggers an alarm and automatically sends SMS with GPS. Optionally call the caregiver number using GSM.
- Location tracking: On demand: Send SMS with coordinates when the button is pressed. Periodic: Send coordinates every N minutes via SMS or via GPRS HTTP POST to a web server where the caregiver app can view the location.

By incorporating these steps, the System can detect objects more accurately under various real-world scenarios. It ensures a safe and independent life for elderly people.

VII. SURVEY OUTCOME

The Study determines strong progress in fall detection and elderly assistance using IoT, AI, cloud-based Technologies, Radar-based tech, vision-based and video systems, wearable sensors, GPS and Navigation Systems. The result achieved accuracy levels above 95%, and a good response to using this stick. Moreover, future research must focus on strength, extensive trials, intuitive design, network security, and unique intelligent systems to make these technologies broadly usable and faithful.

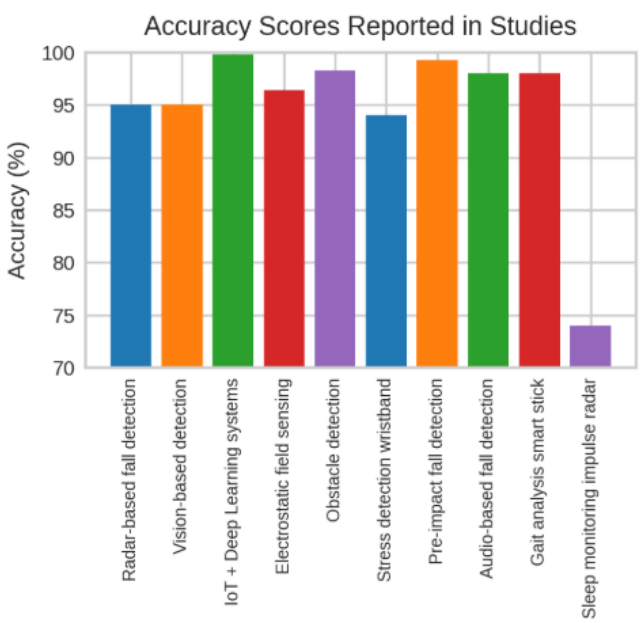

Fig. 2: Model Performance Comparison

A. Radar-Based Technologies

Radar and RF systems are broadly used because they provide non-contact monitoring, work even in dark environments, and maintain Security. The Tech used in this field is Continuous-wave (CW) radar, Ultra-wideband (UWB) radar, Wideband radar models and Electrostatic field sensing.

B. Vision-Based and Video Systems

These systems determine motion patterns visually, helping to detect falls, obstacles, and unusual conditions. It mainly Includes Optical flow, motion-boundary features, Fall Motion Mixture Models, Surveillance cameras, RGB-D cameras, TensorFlow/Viola–Jones object detection, and Neural networks for object classification.

C. Wearable Sensor Technologies

Wearables are more effective for individual, real-time detection and pre-impact fall prediction. It involves Accelerometers, Gyroscopes, Pressure sensors, Wristbands (stress monitoring), Single tri-axial accelerometers for fall prediction and Wearable multi-sensor fusion devices.

D. IoT, AI, and Cloud-Based Systems

IoT enables continuous data collection and remote caregiver alerts. It focuses on AIoT smart sticks, Cloud-assisted monitoring platforms, Smartphone audio analysis for fall detection, and Edge-computing wearable devices

E. GPS & Navigation Systems

These technologies support outdoor safety, navigation, and emergency alerts. It is designed with a Low-energy GPS tracking method, Selective satellite tracking algorithms and GPS-based assistance for elderly individuals.

VIII. ISSUES AND CHALLENGES

A. Reliability Concerns

A lot of migration aids expose problems with sensor precision, battery life and versatile usage for indoor or outdoor scenarios. These dependability issues, in turn, decrease the level of user confidence and make credibility feasible.

B. Usability and Acceptance

A big issue is that several elderly people do not understand or know what to do with these modern assistive devices. With the evolution of new Tech and Development, Many people may not accept the device as user-friendly. By doing all the modern work and all, it will decrease the Usability levels.

C. High cost and Limited Accessibility

The features like GPS tracking, fall detection or analytics with the help of artificial intelligence lead to a higher cost of a device on the whole. Therefore, these technologies remain out of reach for a considerable number of elderly users—particularly in low-income communities where they simply cannot afford them.

D. Maintenance and Privacy Issues

Products that are used daily also need regular maintenance. Systems that track location, activities, or health information over time raise privacy issues. If the data is not handled accurately, they may be misled. A lot of data privacy concerns may be addressed to maintain it effectively.

E. Need for Simpler and User-friendly design

The entire design should be simple and user-friendly to attract a larger number of users in the elderly field. It should be known the exact user to develop a simpler and user-needed system.

IX. CONCLUSION

In this Paper, we came up with a new idea, which helps the elderly people in real life. By integrating Many features like GPS Tracking, SMS Adoption and Mobile Application with user information maintenance, it best suits the elderly's needs and can avoid unusual or emergencies.

The Walking sticks have been traditionally used for longer and served as a simple walking cane, which was normally used by elderly people for basic support and to live independently. These standard traditional devices are limited to just physical assistance and cannot sense dangerous situations, ask for help or communicate during emergency times. Recent advancements in technologies like Internet of Things (IoT) areas, and low-power wireless communication technologies enable and open the door of opportunities in the development of intelligent mobility aids. By integrating sensors and connectivity modules, a smart walking stick can continuously track user movement patterns, detect abnormalities or falls, trigger automated SOS alerts, help in medical support and send real-time location to caregivers or emergency first responders. This goes exceeding the expectations and capabilities of a standard walking cane.

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